

I chose to make the number guessing game as my final project because I thought it would be easy but I wanted to challenge myself and after asking in class if it was okay to make the project into a GUI application I decided to do so. In the game you get to choose three difficulties: Easy, Medium, and Hard. The first one is a range between 1-10 and the next two go up by a power so the guesses can be as high as 1000. When you click on one of the three buttons it takes you to a screen with a textbox that you have to type a number into and you hit the submit button and it tells you if your guess is higher or lower than the randomly generated number. If you type in something that isn't a number the program tells you invalid input. Once you guess the right number the program congratulates you and tells you how many guesses it took you to get to the right answer. Once you get the right answer the submit button is dead and you can't use it anymore so to exit you just click the close button to get out of it. The program also uses colors from the Catppuccin Macchiato color palette because I think that it looks pretty. I personally work in this color scheme and thought it would be nice to share it.

This project I wanted the program to be a GUI application and I quite enjoy PyQt5 so it is heavily object oriented with many functions that link into each other. For the data structure I used a dictionary to store all the potential difficulties because if I didn't do that then I had to write each difficulty by hand which would have sucked. The code looks much cleaner with the setting dictionary rather than a function that does the same thing, it also makes the code much easier to edit. Imagine having to go through and typing each of the CSS values over and over again. The program is split into four different functions `__init__`, `initUI`, `start_game`, and `process_guess`. The `init` initializes all the widgets and their functionality. Without the `init` function then there is no way the buttons or text boxes would work at all. Most PyQt devs tend to have a `initUI` method to keep their code more organized, it's where you store the majority of your CSS that is used later in the program. In this game it is used to customize the buttons and set their functionality. The buttons also use the lambda function to store the variable temporarily from the dictionaries instead of me having to write a loop or write over the buttons entirely. The `start_game` function is there to keep track of the things that are happening in the game, it makes sure everything is initialized properly. The gem of the program is the `process_guess` function which is the core functionality of the number guessing game. I coded this part without a GUI in mind so with a few modifications you can make it terminal based once again but this part of the code is what checks if you inputted the right number or not.

The biggest challenge of this project was translating a terminal based app into a GUI application. I had to relearn PyQt5 because I haven't used it in a while but also some things just don't work when you change what you are designing for. One thing is that you can not use a loop the way the project asks you to do. Originally the `process_guess` function was just a loop but it ended up just freezing the program so I had to turn it into a function so it didn't do that. It still works the same way it would've in a loop and it updates the score just like how it would've if it was a loop. Another challenge was getting an integer value from the text box, originally it was giving me an error for attempting to use it all but I looked up the answer on stackoverflow and found out a way how to properly typecast it. Another challenge was that when you hit the submit

button after you got the correct answer the program would just crash so I had to change it so that once you got the correct answer the program would no longer allow you to hit the submit button.

A possible improvement upon this project would be to add replayability to the program so that once you finish you could have the option to quit the game or to restart. Another thing would be to add an icon to the program but I might just do that in the future when showing it off to other people.